

4.5 Option 3: Life on the Margins: the Food Supply Problem

Life on the margins Significant numbers of people live a life on the margin, in a situation of food insecurity. Others consume more than their fair share of global resources. This option explores this inequality, focusing on regions where food production is a continual challenge. The 'margins' may be traditional areas of famine, but also rapidly urbanising areas where food is scarce and malnutrition an ever present threat.

Food insecurity The causes of food insecurity are complex, ranging from physical processes of land degradation and desertification, exacerbated by human over-exploitation, to population pressure and political processes. Increasing food supply represents a key challenge cemented with the Millennium Development Goals. However, often agreement does not exist on the best way to achieve this; options range from reforming trade systems to hi-tech farming, intermediate technology and organic farming.

Global synoptic context

Key linkages

Places

Where are the 'margins'?

People

Who suffers from food insecurity and over-nutrition?

Power

Which organisations and groups are involved in ensuring food security?

Suggested teaching and learning

Mapping the distribution of countries suffering from food insecurity, and those facing the obesity threat.

Contrasting the socio-economic profiles of countries at different end of the food supply spectrum.

Briefly investigating the role of UN bodies such as the FAO, and NGOs such as Practical Action.

1 Global and local feast or famine

Enquiry question: *What are the characteristics of food supply and security?*

What students need to learn	Suggested teaching and learning
■ Current issues associated with food supply and security; such as food miles, famine, globalisation of food tastes, under and over nutrition.	■ Exploring the current socio-economic issues associated with feeding people globally and locally.
■ The environmental issues resulting from food production.	■ Developing an awareness of inappropriate farming techniques and their impact on the environment.
■ Why food supply varies spatially.	■ Investigate how current patterns of food supply and security vary at different scales in rural and urban environments.
■ What life on the margins means to differing people and the role of food security in the overall quality of life there.	■ Developing an awareness that life on the margins of survival ranges from traditionally perceived areas of famine such as the Sudan to megacities where poverty has impacts on food supply and consumption.

2 The complex causes of food supply inequalities

Enquiry question: *What has caused global inequalities in food supply and security?*

What students need to learn	Suggested teaching and learning
■ The complex interlinking and overlapping causes of famine and food surpluses and classification of the environmental, social and economic causes into long- and short-term, direct and indirect causes.	■ Investigating how and why areas of food surplus and famine have emerged.
■ The role of population pressure in creating food insecurity.	■ Weighing up the contrasting viewpoints on population pressure as highlighted by neo-Malthusians versus technocentric followers of Boserup.
■ The impacts of attempts to increase global food supply.	■ Researching food production techniques which have been environmentally damaging and contrasting these with more environmentally friendly production methods such as LEAF and organic schemes in locations as diverse as the UK, Cuba, China and Africa.
■ Who has been most affected by food insecurity, and why.	■ Investigating the nature of groups of people vulnerable to food insecurity.

3 Desertification and life at the margin of survival

Enquiry question: *What is the role of desertification in threatening life at the margins?*

What students need to learn	Suggested teaching and learning
■ Desertification and its scale and impact.	■ Developing an awareness that desertification is a type of land degradation and investigating the varying human activities and climatic variations causing it.

■ The scale and characteristics and vulnerability of dryland ecosystems.	■ Investigating the specific ecosystems, climate and hydrology of drylands to establish the physical causes of risk and vulnerability.
■ Why drylands are extremely vulnerable to over-exploitation and inappropriate land use.	■ Categorising and investigating the human factors causing vulnerability (socio-political and economic).
■ The relationship between food production and supply in desertified regions.	■ Researching how food supply is affected by desertification and conversely how food supply methods can cause desertification such as salinisation.

4 The role of management in food supply and security

Enquiry question: *How effective can management strategies be in sustaining life at the margins?*

What students need to learn	Suggested teaching and learning
■ Management techniques and strategies that have attempted to increase global food supply and security.	■ Researching why management strategies are not just focused on food production but fair trade and distribution and even the politics and stability of an area.
■ The sources of these developments and why greater international efforts are increasingly needed.	■ Developing an awareness that several alternative strategies exist for providing secure global food supplies; some are technology-dependent, others are low-cost (eg intermediate technology) while others may be focused upon the environment eg, organic farming.

■ Initiatives that have been most effective in sustaining life at the margins.	■ Weighing up the range and role of organisations involved in marginal food supply areas (government, business, NGOs and increasingly international organisations such as the United Nations Environment Programme).
■ The role of sustainable strategies in food supply and security.	■ Investigating a range of strategies that are designed to be sustainable, such as assessing the potential of fair trade, organic farming or of the aquaculture in sustaining life at the margins of society.

Suggested fieldwork opportunities for life on the margins: the food supply problem

Local fieldwork ■ Investigating issues of sourcing of food including local sourcing eg for supermarkets, farm shops and catering establishments. ■ Visits to contrasting local farms including organic, LEAF schemes. ■ Fair trade surveys.	Residential and long-haul fieldwork ■ Areas such as Morocco can provide studies of life at the margin and desertification. ■ American west could provide studies of dry land management.
--	--